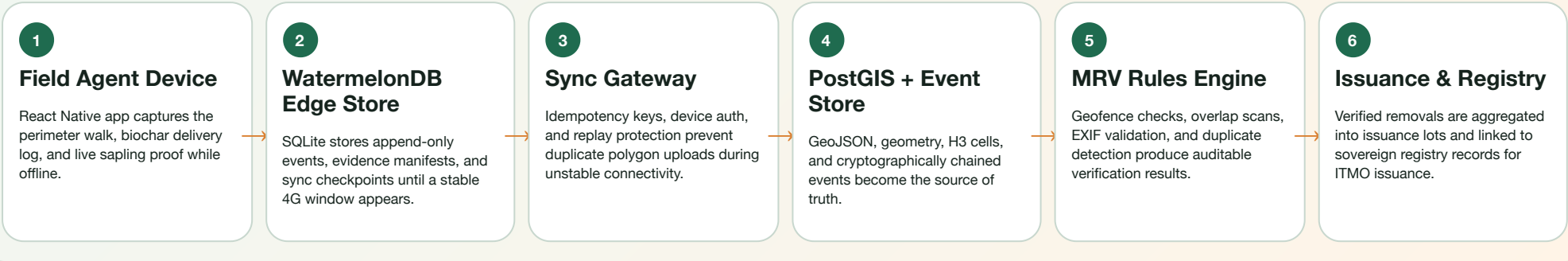


Distributed Architecture & Spatial Schema

India-scale MRV foundation for offline capture, fraud resistance, and zero double-counting under Article 6.2.

EDGE TO REGISTRY FLOW



farmers

PK farmer_id uuid
name text
adv_reference_token text unique
phone_e164 text
upi_vpa text
preferred_language text
consent_status text
created_at timestampz

farm_polygons

PK polygon_id uuid
FK farmer_id uuid
geojson jsonb
geom geometry(Polygon,4326)
boundary_hash text
area_hectare numeric(10,4)
survey_accuracy_m numeric(6,2)
created_at timestampz

polygon_h3_cells

PK polygon_id uuid
PK h3_cell text
resolution smallint
coverage_ratio numeric(6,4)

evidence_assets

PK asset_id uuid
capture_mode text
file_sha256 text
exif_json jsonb
capture_signature text
captured_at timestampz
captured_geom geometry(Point,4326)
inside_geofence boolean

biochar_ledger

PK event_id uuid
FK polygon_id uuid
FK evidence_asset_id uuid
quantity_kg numeric(10,2)
application_method text
applied_at timestampz
capture_point geometry(Point,4326)
status text

tree_inventory

PK tree_id uuid
FK polygon_id uuid
FK evidence_asset_id uuid
species text
planting_date date
capture_point geometry(Point,4326)
height_cm numeric(8,2)
status text

CORE RELATIONSHIPS

RELATION	PURPOSE
farmers 1:N farm_polygons	Each farmer can hold multiple mapped plots across fragmented landholdings.
farm_polygons 1:N biochar_ledger	Every biochar application is anchored to a verified polygon and a geofenced capture point.
farm_polygons 1:N tree_inventory	Tree records remain parcel-specific for future survival, canopy, and sequestration checks.
farm_polygons 1:N polygon_h3_cells	H3 cells accelerate overlap detection and portfolio-level heatmaps without expensive polygon scans.
evidence_assets linked to ledger + trees	Photos, signatures, and EXIF proofs are re-usable evidence objects instead of opaque blobs in business tables.

APPEND-ONLY EVENT STORE

event_store

PK event_store_id uuid
aggregate_type text
aggregate_id uuid
event_type text
payload jsonb
prev_hash text
curr_hash text
device_signature text
recorded_at timestampz

Tamper-proof chain: `curr_hash = SHA-256(payload || recorded_at || prev_hash)`.
Any change to a past event breaks the hash chain and becomes immediately visible to auditors.

All farmer, polygon, biochar, tree, and payout events are written here first and projected into read models for dashboards and registry exports.

Aadhaar data is not stored in Joita databases. The platform keeps only an `adv_reference_token` issued by a certified Aadhaar Data Vault, which keeps the design aligned with India's DPDP posture.